

Abhi Naglapura

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SUMMARY

Third-year Mechatronic Engineering student (Honours) at UTS with hands-on experience across embedded systems, PCB design, robotics, and aerospace. Leads competition rocketry and embedded hardware end-to-end - firmware, custom PCBs, control, and mechanical integration, with a track record of delivering working systems and a focus on requirements, integration, and verification.

EDUCATION

Bachelor of Engineering (Honours), Major in Mechatronic Engineering

Jan 2024 – Dec 2028 (expected)

University of Technology Sydney (UTS)

- High Distinctions: Introduction to Mechatronics Engineering; Embedded Mechatronics Studio
- Distinctions: Sensors and Control (83); Programming for Mechatronic Systems (83); Mathematics 2 (77); Industrial Robotics; Introduction to Mechanical Engineering
- Coursework in progress: Space Robotics; Space Engineering Design Studio; Robotics Studio 1

NSW Higher School Certificate

2018 – 2023

The Ponds High School

- Ranked 1st in school cohort: HSC Engineering Studies (87%)

TECHNICAL SKILLS

Languages: C, C++, Python, MATLAB / Simulink, Java, G-code

Embedded & Hardware: STM32 (CubeIDE, HAL), ESP32, Arduino, ATtiny; PCB design (Altium, KiCad, JLCPCB), 4-layer boards; CAN bus, SPI / I²C; analog filter design

Robotics & Vision: ROS, ROS2, Gazebo, OpenCV, image-based visual servoing (IBVS), Resolved Motion Rate Control (RMRC), Docker

Systems & Methods: Requirements, subsystem integration, verification & validation (V&V), design reviews, OpenRocket simulation

CAD & Manufacturing: SolidWorks, Fusion 360, Onshape, ANSYS Fluent (basic), Quartus Prime; 3D printing (FDM), CNC & manual machining, laser cutting

Tools & OS: Git, Linux & embedded Linux (Raspberry Pi), VS Code

RELEVANT EXPERIENCE

Project Lead - Project Shingleback

Nov 2025 – Present

UTS Rocketry

- Lead end-to-end development of a 5.5" fibreglass rocket to 10,000 ft (AURC 2026, COTS), coordinating requirements, design, and integration across seven subsystems
- Define subsystem interfaces and a verification & validation (V&V) matrix; own master OpenRocket flight simulations
- Direct development of an SRAD LQR airbrake (apogee control), a CAN-bus avionics stack, and a 3U CubeSat payload (six MPU6050 IMUs on an ESP32-S3)
- Run design reviews, risk assessments, and feasibility analysis to keep subsystems on schedule

Electrical Lead - Embedded Mechatronics Studio

2026

University of Technology Sydney

- Own the STM32G474 microcontroller and system-peripheral subcircuits (USB-C, SWD, FPC, user I/O) on a 4-layer JLCPCB board
- Design an ADXL335 accelerometer front-end with a Sallen–Key Butterworth low-pass filter and a bq21040 charger / TPS2113 power-mux supply
- Debug board-level faults through to delivery and implement a static-acceleration calibration routine for step detection

Launch Officer

Jan 2025 – Dec 2025

UTS Rocketry

- Coordinated launch-day operations — motor-box setup, black-powder ignition, and recovery
- Enforced range-safety procedures and led the launch team through high-pressure flight operations

Internal Structures Lead - Project Goanna

Jan 2025 – Sep 2025

UTS Rocketry

- Designed and fabricated lightweight load-bearing rocket structures, balancing strength against mass and integration
- Coordinated mechanical integration of propulsion, electronics, and recovery subsystems

PROJECTS

Visual Servoing on a UR3 Arm

Sep – Oct 2025

Sensors and Control (41014), UTS

- Built an eye-in-hand image-based visual servoing (IBVS) system driving a physical UR3 to centre an ArUco marker
- Computed corrective joint velocities from real-time camera feedback using OpenCV, running ROS Noetic in Docker

Collaborative Multi-Arm Assembly

Sep – Oct 2025

Industrial Robotics (40103), UTS

- Simulated four coordinated robot arms (ABB IRB120, UR3, myCobot) assembling a structure using Resolved Motion Rate Control
- Built an ImGui control panel and integrated a hardware E-stop over serial via Arduino for fail-safe shutdown

Multi-Platform Autonomous Navigation (ROS2 / C++)

2026

Programming for Mechatronic Systems, UTS

- Built a C++ / ROS2 autonomy stack for Ackermann and skid-steer platforms with laser and sonar sensing in Gazebo
- Implemented a mission coordinator and obstacle detection spanning multiple vehicle types

Automated Satellite Tracker

Jul – Nov 2024

Introduction to Mechatronics (41099), UTS

- Built an azimuth/elevation antenna tracker driving stepper motors from live Orbitron satellite data over serial
- Targeted amateur-radio satellite reception and UAV ground-station applications

OTHER EXPERIENCE

Volunteer Staff & Supply Officer

Mar 2024 – Present

Australian Air Force Cadets — 320 Squadron

- Deliver aviation theory and technical training to youth cadets, supporting their leadership and teamwork development
- Manage logistics and supply for 50+ cadets, including procurement, stock control, and uniform distribution

Private Tutor (STEM)

Dec 2023 – Present

Alchemy Tuition

- Deliver tailored one-on-one maths, science, and engineering tutoring, improving student results by 20%+

Lifeguard

Oct 2025 – Present

Raging Waters Sydney

- Maintain safety supervision and rapid incident response across a high-traffic aquatic facility

REFEREES

Available on request.